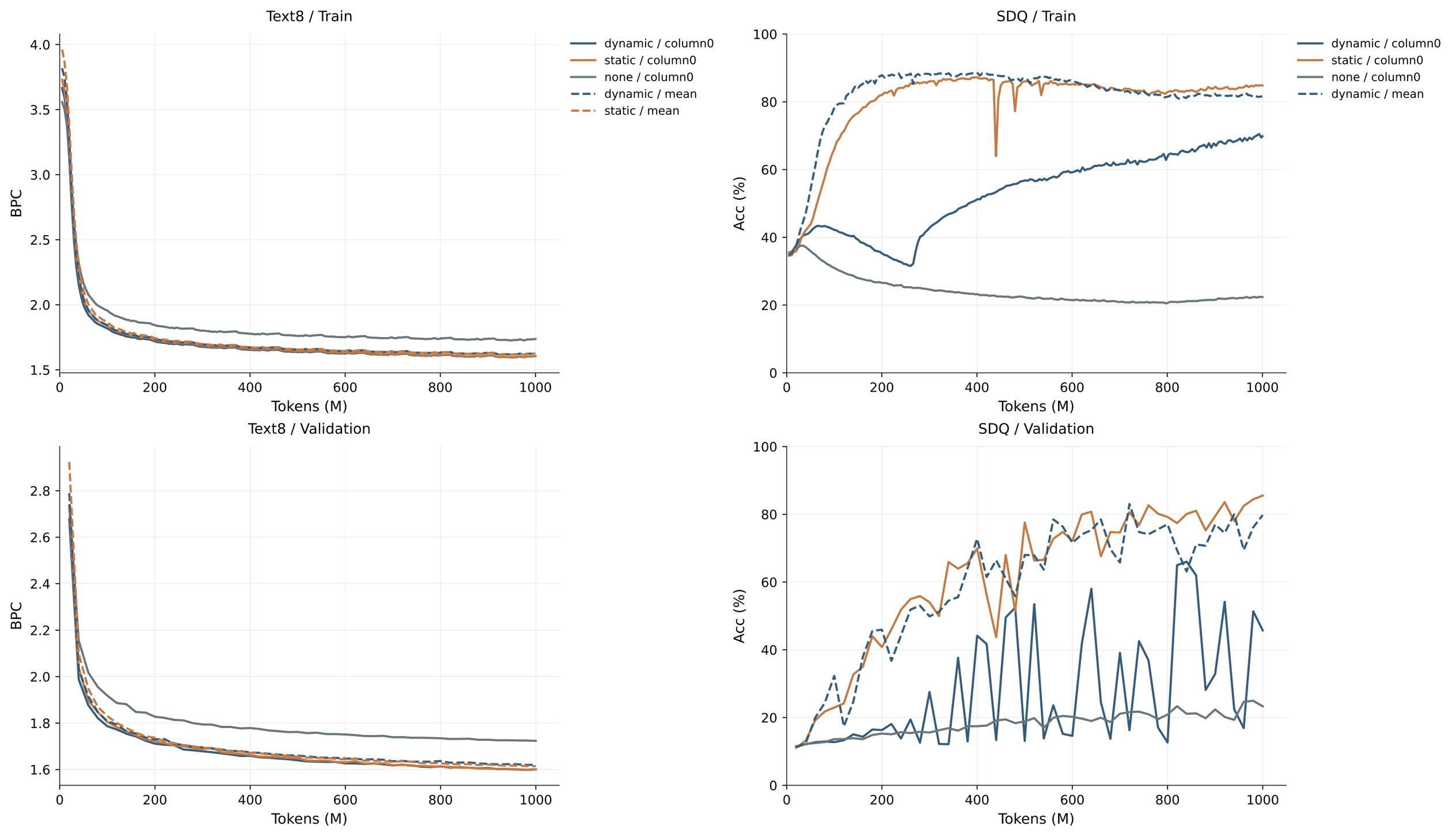
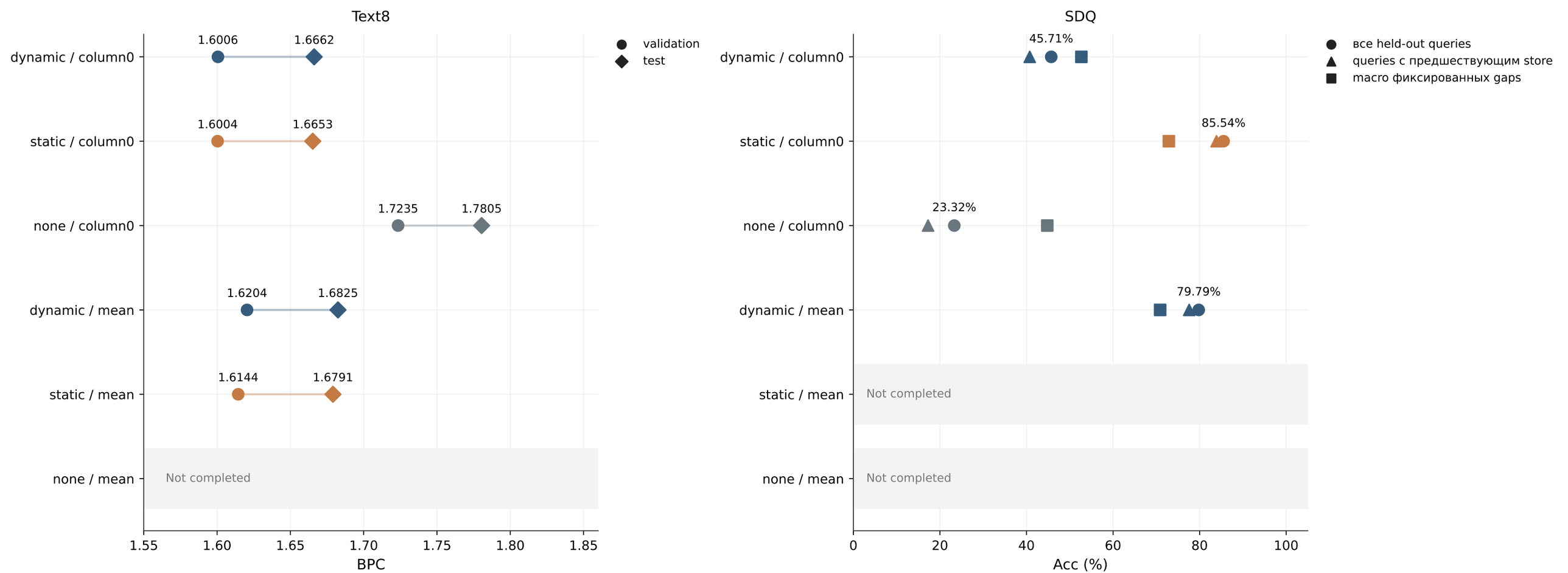
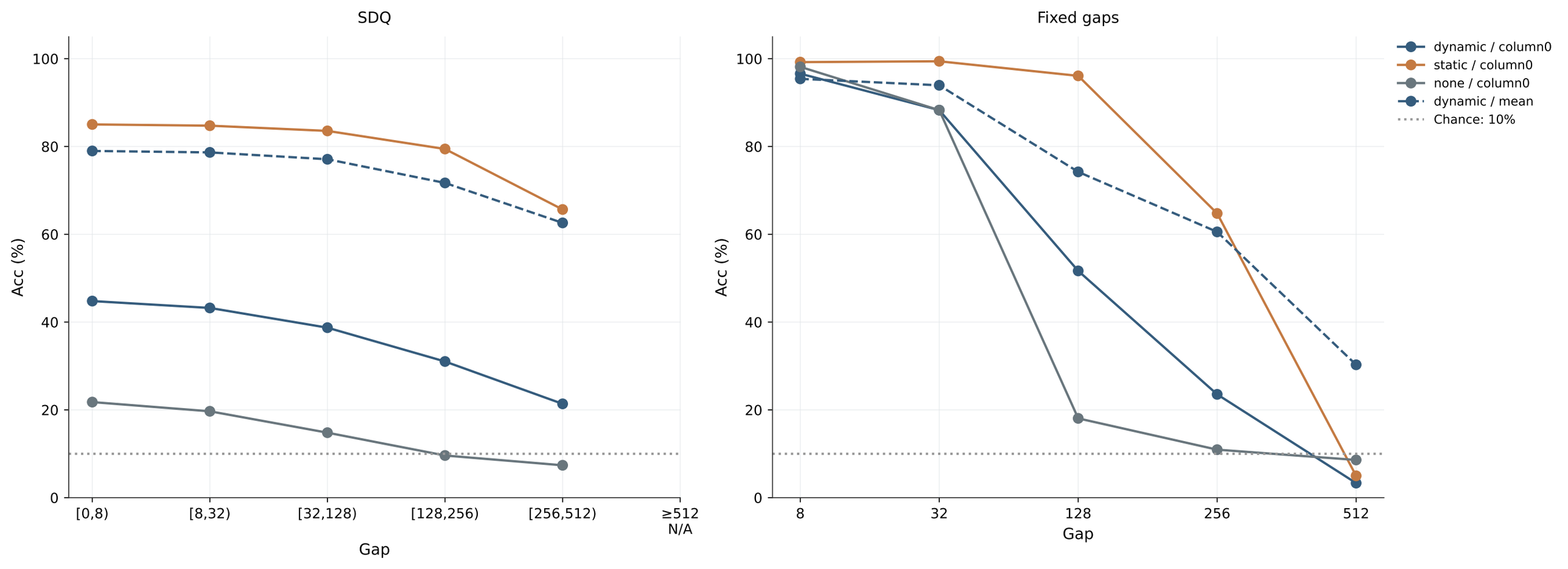
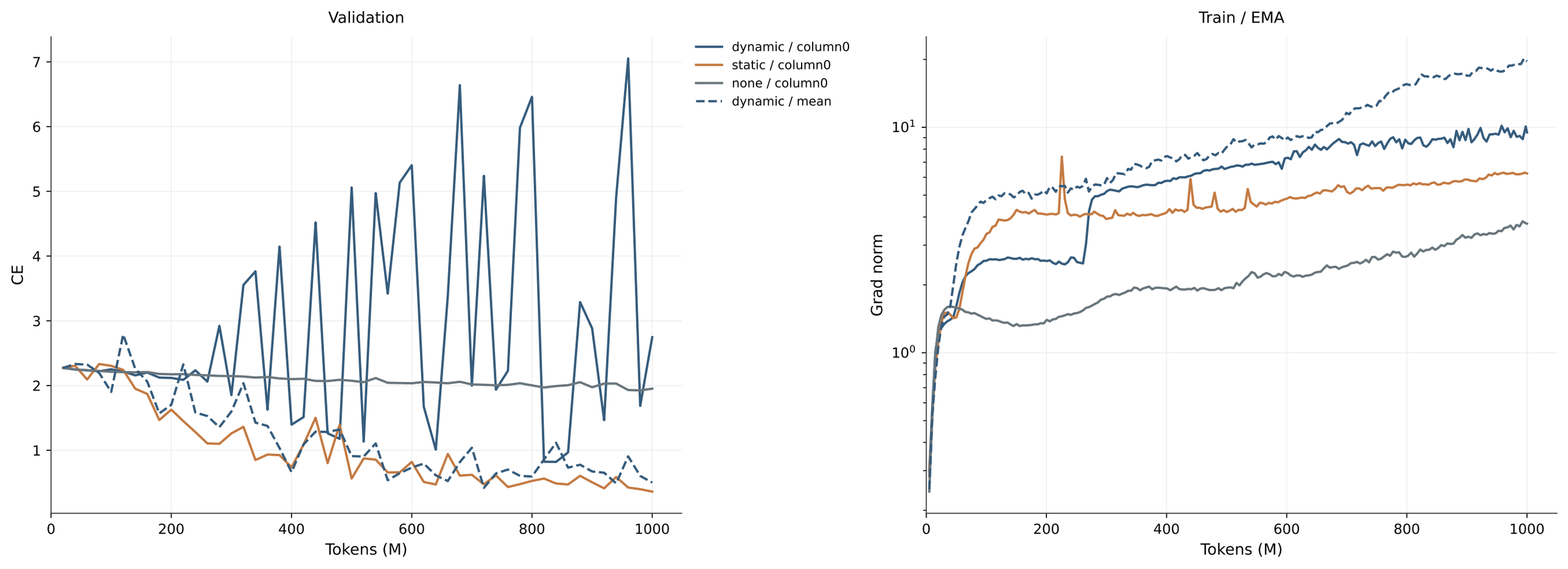
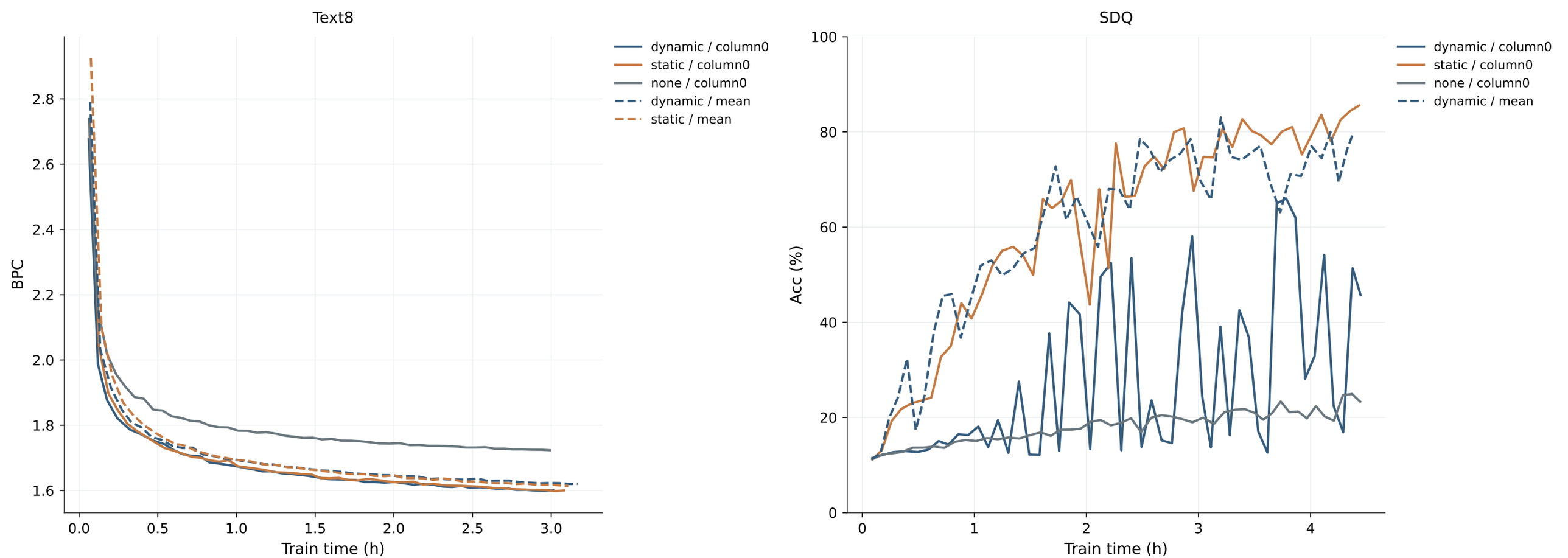
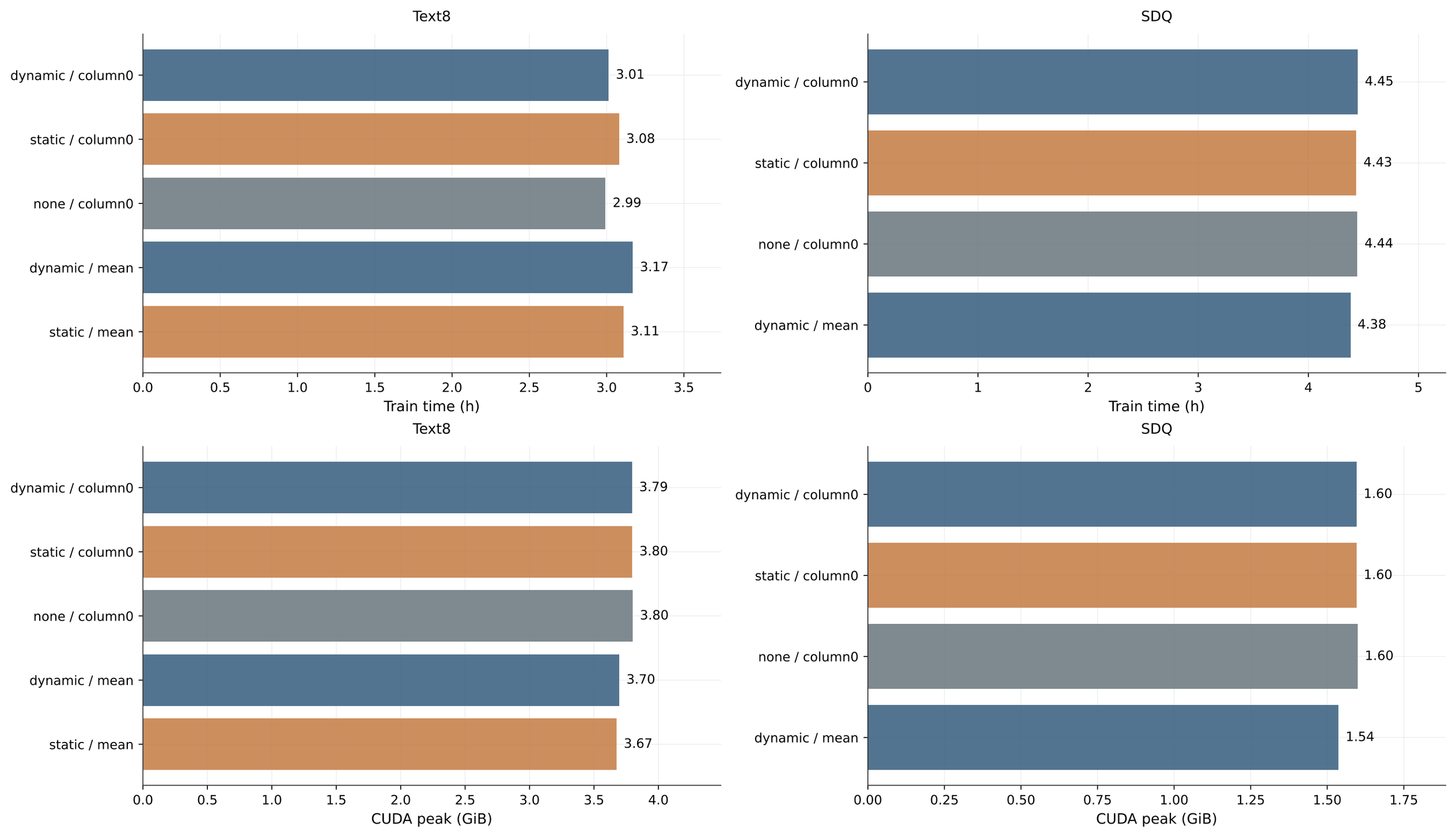

Text8 / Train
BPC
Tokens (M)
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
SDQ / Train
Acc (%)
Tokens (M)
dynamic / column0
static / column0
none / column0
dynamic / mean
Text8 / Validation
BPC
Tokens (M)
SDQ / Validation
Acc (%)
Tokens (M)

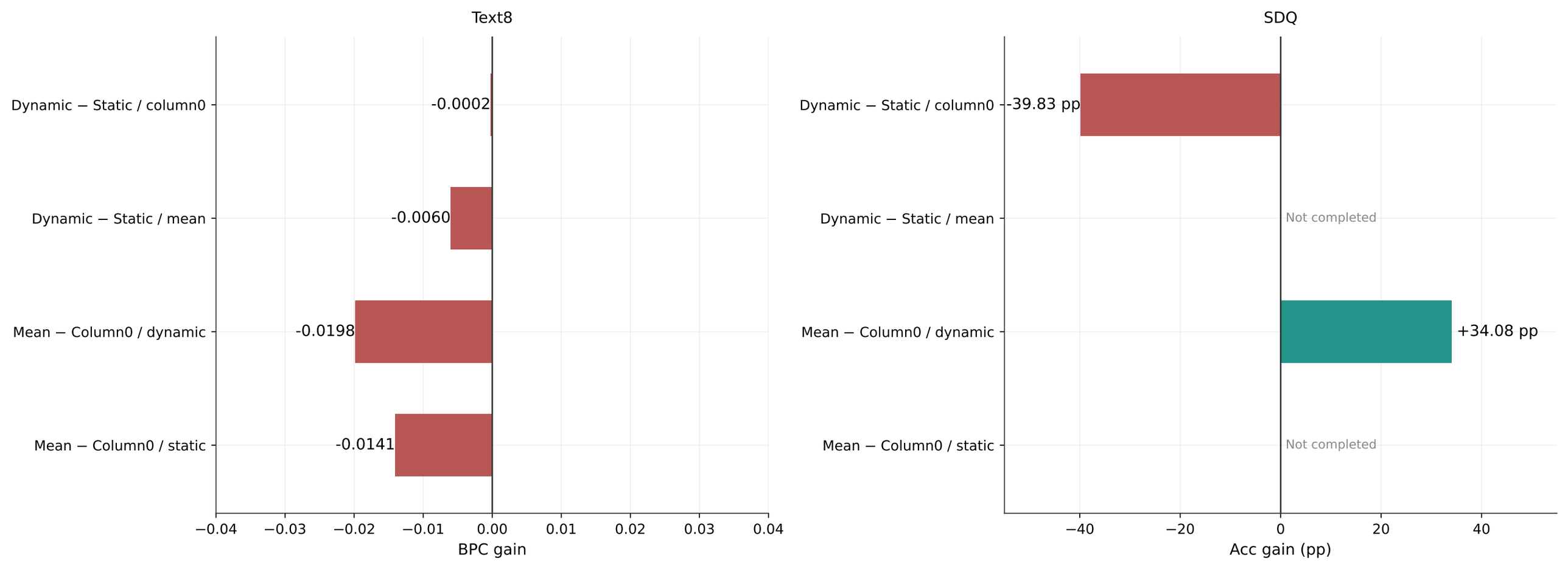

Text8
validation
test
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
none / mean
1.6006
1.6662
1.6004
1.6653
1.7235
1.7805
1.6204
1.6825
1.6144
1.6791
Not completed
1.55
1.60
1.65
1.70
1.75
1.80
1.85
BPC
SDQ
все held-out queries
queries с предшествующим store
macro фиксированных gaps
45.71%
85.54%
23.32%
79.79%
Not completed
Not completed
0
20
40
60
80
100
Acc (%)


SDQ
Fixed gaps
Acc (%)
Gap
[0,8)
[8,32)
[32,128)
[128,256)
[256,512)
≥512
N/A
8
32
128
256
512
dynamic / column0
static / column0
none / column0
dynamic / mean
Chance: 10%


Validation
CE
Tokens (M)
dynamic / column0
static / column0
none / column0
dynamic / mean
Train / EMA
Grad norm
Tokens (M)


Text8
BPC
Train time (h)
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
SDQ
Acc (%)
Train time (h)
dynamic / column0
static / column0
none / column0
dynamic / mean


Text8
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
3.01
3.08
2.99
3.17
3.11
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Train time (h)
SDQ
dynamic / column0
static / column0
none / column0
dynamic / mean
4.45
4.43
4.44
4.38
0
1
2
3
4
5
Train time (h)
Text8
dynamic / column0
static / column0
none / column0
dynamic / mean
static / mean
3.79
3.80
3.80
3.70
3.67
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
CUDA peak (GiB)
SDQ
dynamic / column0
static / column0
none / column0
dynamic / mean
1.60
1.60
1.60
1.54
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
CUDA peak (GiB)


Text8
Dynamic − Static / column0
Dynamic − Static / mean
Mean − Column0 / dynamic
Mean − Column0 / static
-0.0002
-0.0060
-0.0198
-0.0141
−0.04
−0.03
−0.02
−0.01
0.00
0.01
0.02
0.03
0.04
BPC gain
SDQ
Dynamic − Static / column0
Dynamic − Static / mean
Mean − Column0 / dynamic
Mean − Column0 / static
-39.83 pp
Not completed
+34.08 pp
Not completed
−40
−20
0
20
40
Acc gain (pp)